

Reduced Order Modeling (POD-ROM) for 2D Heat Equation

This project presents a POD-based reduced order model for the 2D heat equation using SVD.

Governing Equation: $dT/dt = \alpha * (d^2T/dx^2 + d^2T/dy^2)$

Mode	Energy	Cumulative Energy
1	0.974933	0.974933
2	0.024476	0.999408
3	0.000584	0.999992
4	7e-06	1.0
5	0.0	1.0
6	0.0	1.0
7	0.0	1.0
8	0.0	1.0
9	0.0	1.0
10	0.0	1.0
11	0.0	1.0
12	0.0	1.0
13	0.0	1.0
14	0.0	1.0
15	0.0	1.0
16	0.0	1.0
17	0.0	1.0
18	0.0	1.0
19	0.0	1.0
20	0.0	1.0